Phil. Trans. R. Soc. Lond. B **327**, 463–473 (1990)
Printed in Great Britain

The logic of failure

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Unlike other living creatures, humans can adapt to uncertainty. They can form hypotheses about situations marked by uncertainty and can anticipate their actions by planning. They can expect the unexpected and take precautions against it.

In numerous experiments, we have investigated the manner in which humans deal with these demands. In these experiments, we used computer simulated scenarios representing, for example, a small town, ecological or economic systems or political systems such as a Third World country. Within these computer-simulated scenarios, the subjects had to look for information, plan actions, form hypotheses, etc.

1. Introduction

(*a*) *A famine in West Africa*

Our World, unfortunately, is not exactly bereft of catastrophes. Three years ago, the explosion of the Chernobyl reactor made for a great deal of anxiety and a great many discussions, which are still going on, regarding the extent to which humanity can afford technology, the failure of which has such far-reaching consequences.

Questions like these are, of course, especially salient when, as in the case with the Chernobyl disaster, one is dealing with 'man made' catastrophes rather than natural ones. This case, after all, poses the question of avoidability in an especially urgent manner.

The catastrophe I shall be concerned with now, occurred in West Africa, in the Moro region in the state of Bukina Faso which is shown in figure 1. That catastrophe hardly made any headlines. Twelve years ago a development aid project was begun there with the aim of creating generally better living conditions for the semi-nomadic people living in this region. By sinking deep wells, the available water supply was increased, the pasture areas for cattle were better irrigated, resulting in a sharp increase in the stock of herds of cattle. In addition, the tsetse fly, which had severely infected the cattle with sleeping sickness, had been brought under control. So the Moros ten years later were considerably better off. The development of some important variables of the Moro region can be seen in figure 2.

In the end, herds of cattle exceeded the capacity of the available pasture area; a slight decrease in rainfall led to a food shortage; the pastures were overgrazed, and the hungry cattle destroyed the turf and hence the basis for vegetation: the cattle starved to death, thus giving rise to a famine in the human population, which claimed many lives.

Fortunately, this catastrophe did not actually take place in West Africa, but in an office in downtown Zurich. The cause of the catastrophe was not due to a team of development aid volunteers but an executive of an international computer enterprise. The Moro region merely existed in the form of a computer program and serves us as an instrument for investigating the action strategies implemented by persons in coping with extremely complex, dynamic, intransparent and uncertain systems. I shall now describe these attributes of a system more closely.

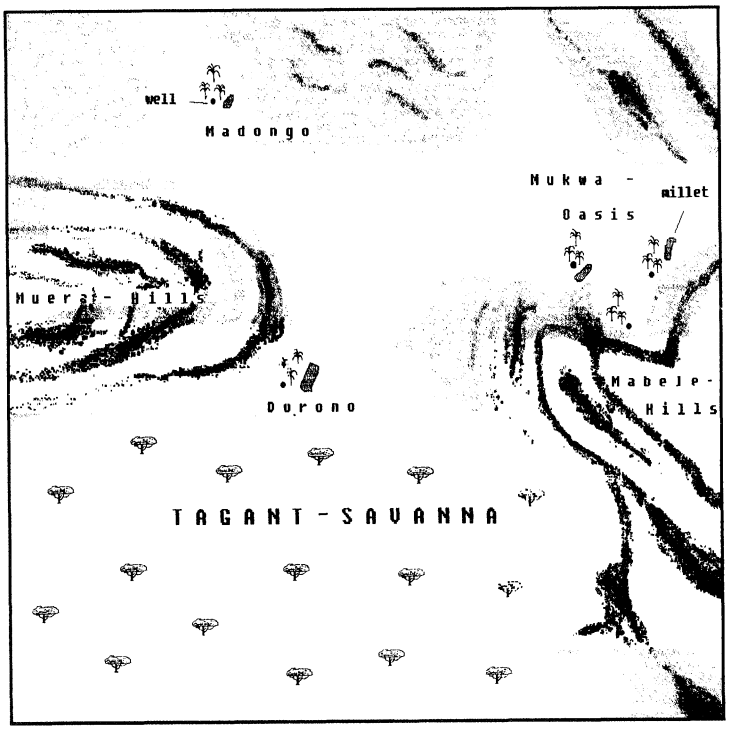


FIGURE 1. The Moro region in Bukina Faso, West Africa.

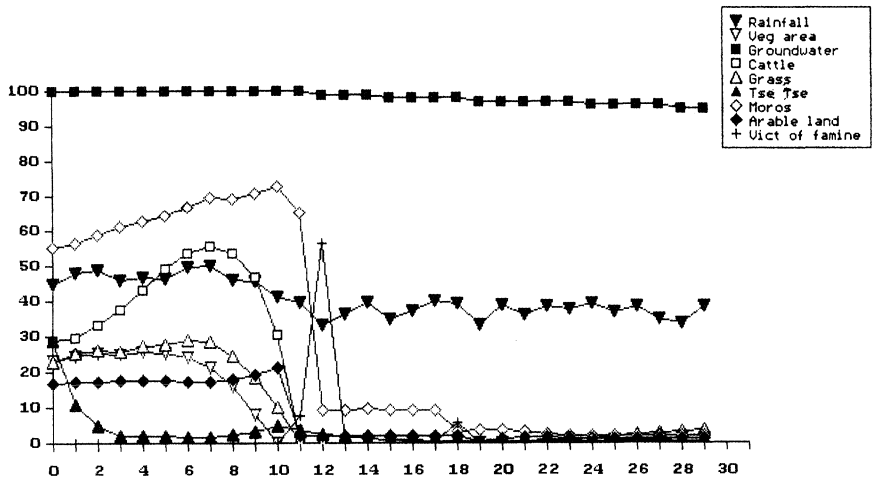


FIGURE 2. A famine in the Moro region.

A system is extremely complex when it consists of a great variety of variables. The Moro system consists of the following variables: population with birth-rate and mortality, cattle stock, ground water, vegetation area, precipitation, area of arable land, millet harvest, etc. All these variables are closely tied to one another, they mutually affect each other and constitute a network of interdependencies, which are shown in figure 3.

The Moro system is dynamic, which means that it develops further, even without interventions. It does not remain stable and waits for interventions like, for instance, a chess board. Because the Moro system is intransparent, many of the variables, for example, the

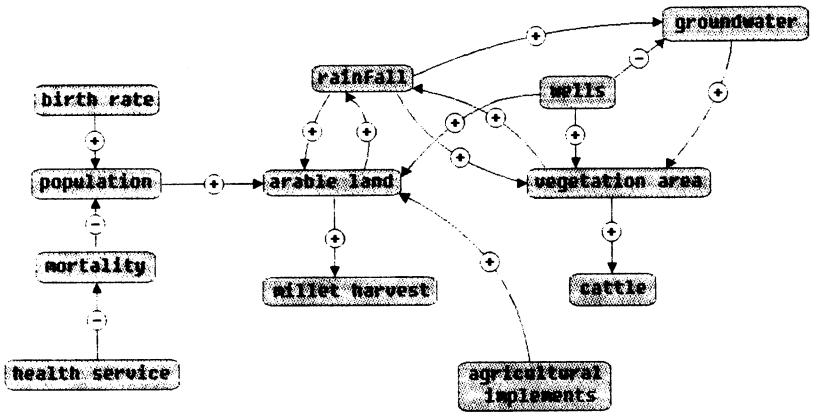


FIGURE 3. Part of the network structure of the Moro system.

groundwater level, defy direct observation. The uncertainty of the Moro system means that the acting subject has no complete knowledge about the system, about its variables and their interdependencies.

For our experimental subjects, the Moro system is a 'dynamic decision problem' (Brehmer & Allard 1986) as the experimental subjects are included in the development of the system. They gather information at a certain point in time, make decisions for particular measures and then execute these measures; they are confronted with the effects of their decisions, are able to gather further data, decide for the same or new measures, and so on. Figure 4 shows the results of the governmental activities of a good subject. It is possible to create positive living conditions for the Moros over a long period of time.

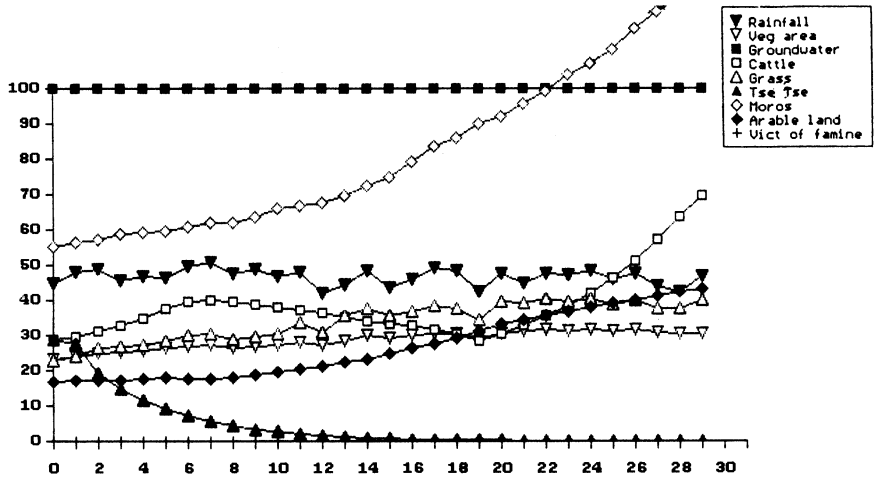


FIGURE 4. Successful governmental activity in the Moro region.

Of course, we do not investigate the behaviour of our experimental subjects in the Moro region with the intention of learning something about the action strategies of development aid volunteers in West Africa. Instead, we believe that action situations, such as those in the Moro region, are in many respects typical of the conditions for action in the rest of the world.

The reason for our experiments with computer-simulated environments is to gain a general insight into the psychology of action-regulation in complex fields of reality. People have to meet

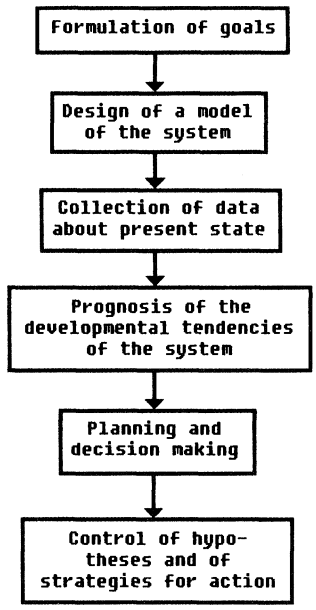


FIGURE 5. Demands of complex action regulation.

various demands in such a situation, which are not at all typical only for the Moro system. Figure 5 shows the tasks that must be performed in coping with such a system.

The first important task is that of formulating goals. More often than not, one has to first get the goals clear in one's mind, as the goals in complex action situations are often given in an indistinct form. One must posit hypotheses on the inner structure of the system with which one is coping. Information must be gathered on the current state of the system at any given time so as to predict the system's developmental trends. Plans and decisions have to be made, and hypotheses and action strategies must be tested. All of these diverse activities must be executed in a manner appropriate to whatever situation is being coped with. When under time pressure, for instance, one should not only plan within a shorter period of time, but differently than when there is plenty of time available. This is discussed later.

2. THE REASONS FOR SUCCESS AND FAILURE

What, then, is behind the success or failure in complex action tasks? How do people meet the various demands in a complex action situation of this kind? First of all, differently! In an experiment conducted by Schaub & Strohschneider (1989), we compared the behaviour of 45 executives from large German and Swiss industrial and commercial enterprises with the behaviour of students. Each subject had to try, over a (simulated) period of over 20 years, to improve the living conditions of the Moros. In each year, the subjects were able to ask questions about the (then) current state of the Moros and to make any decisions they wished to. They could buy tractors for the Moros, sink wells, improve medical care, influence trade relations and so on. Figure 6 shows the average results for the executives and the students. On the whole, the executives did far better; they earned, by skilful management, more capital, their herds of cattle were greater, as were their vegetation areas and human populations. There were, accordingly, fewer catastrophes in their simulations.

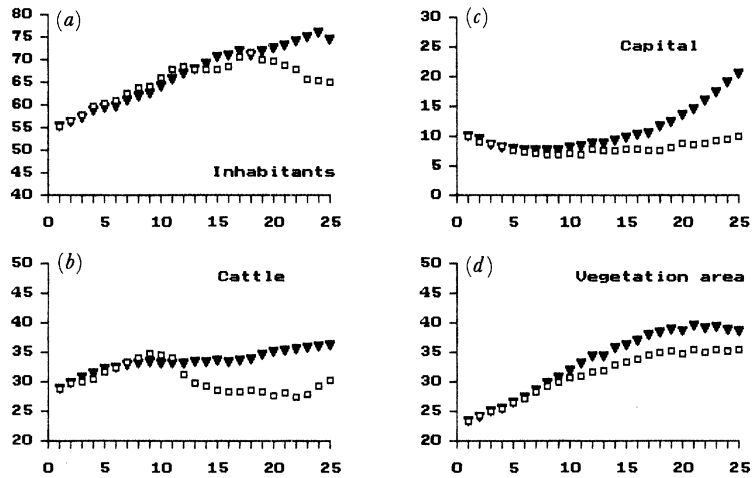


FIGURE 6. The average developments in (*a*), population; (*b*), cattle; (*c*), capital; and (*d*), vegetation, for executives (triangles) and students (squares).

Which mistakes do people make when faced with coordinating complex actions? What are the reasons for catastrophes like the Moro catastrophe described above? We found a great variety of modes of faulty behaviour that were responsible for the failure when acting in such complex situations. Figure 7 presents a rough overview of the various forms of faulty behaviour. Subjects fail, for instance, to get their goals clear and then act according to a 'repair service policy'. They do this by eliminating the obvious errors and solving the conspicuous problems, while disregarding the less conspicuous ones and, of course, failing to take into account aberrant developments that first become apparent in faint symptoms. Subjects fail to construct

insufficient goal elaboration
→ acting according to a 'repair-shop' principle

insufficient formation of hypotheses about the structure of the system
→ neglecting side- and long-term effects

insufficient ideas about the behaviour of the system in time
→ neglecting developmental tendencies of the system

insufficient coordination of different measures
→ 'collision' of measures

'ballistic' action
→ no detection of wrong hypotheses and inappropriate strategies

no self-reflection
→ no 'repair' of wrong hypotheses and inappropriate strategies

FIGURE 7. Modes of faulty behaviour in coping with complex systems.